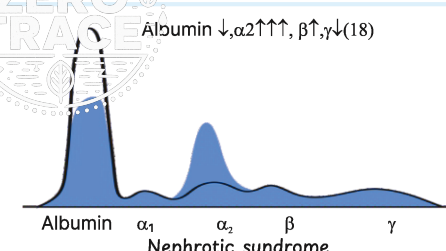
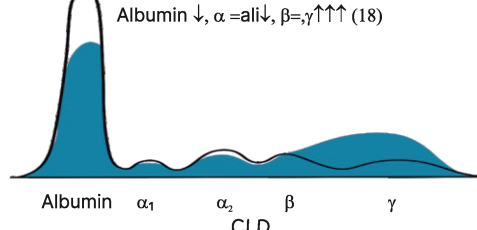
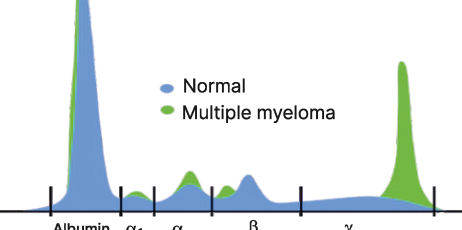


SERUM ELECTROPHORESIS:

 Albumin ↓, α₂ ↑↑↑, β ↑, γ ↓ (18) Nephrotic syndrome	• Nephrotic syndrome ○ Albumin ↓ as it is lost in urine ○ Inflammatory changes present ○ Alpha 2 globulin ↑
 Albumin ↓, α = all ↓, β =, γ ↑↑↑ (18) CLD	• Chronic liver diseases ○ Albumin ↓ ○ Gamma globulin ↑ as globulin is catabolised by liver
 Normal Multiple myeloma Multiple Myeloma	• Multiple myeloma ○ Church spire appearance

CAUSES OF CHRONIC LIVER DISEASES/CIRRHOSIS

- Non-alcoholic fatty liver disease
- Alcoholic liver diseases
- Chronic viral hepatitis B, C, D
- Autoimmune hepatitis: Interface hepatitis, Rosette formation, Anti-LKM1 antibodies
- Primary biliary cirrhosis: Anti-mitochondrial antibodies
- Primary sclerosing cholangitis: p-ANCA antibodies
- Wilson and hemochromatosis
 - Wilson: ATP7B gene, chromosome 13
 - Primary hemochromatosis: HFE gene, chromosome 6
- Alpha 1 antitrypsin deficiency
 - Associated with Panacinar variety of Emphysema
- Cardiac cirrhosis (Nutmeg liver)
- Cryptogenic cirrhosis

CARDIAC CIRRHOSIS

- Cigarette smoker → COPD → Pulmonary artery hypertension → vasoconstriction in pulmonary artery
- Pressure in the pulmonary artery ↑
- Right ventricular congestion → right atrial congestion → ↑ hepatic venous pressure via inferior vena cava
- Initially enlargement of liver → fibrosis by hepatic stellate cells → subsequent regression in liver size



Nutmeg liver

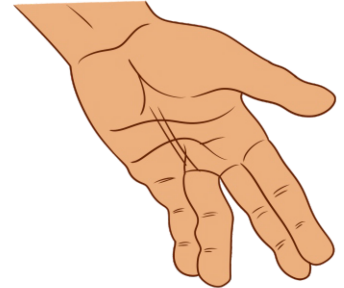
FINDINGS IN COMPENSATED CIRRHOSIS

00:20:08

- **Spider naevi or spider angioma**
 - Superior vena cava distribution
 - Dilated cutaneous arterioles
 - Also seen in SLE, RA, Pregnancy, alcoholic liver diseases
 - Estrogen excess → vasodilation (hormone responsible)
- **Palmar erythema**
- Dupuytren's contracture: Contracture of **ring finger**
- **Clubbing**
 - Fastest clubbing present with lung abscess
- **Parotid** enlargement
- Testicular atrophy (less LH from HPA and ↑ aromatisation to estrogen)
 - Erectile dysfunction
 - ↓ **Body hair** and gynecomastia



Clubbing



Palmar erythema

	COMPENSATED CIRRHOSIS	DECOMPENSATED CIRRHOSIS		
Stage	Stage 1	Stage 2	Stage 3	Stage 4
Clinical	No Varices No Ascites	Nonbleeding Varices + No Ascites	Ascites +/- Bleeding Varices +	Encephalopathy Bleeding +/- Ascites +
Death (at 1 year)	1%	3%	20%	57%

- Nonbleeding varices treatment: Endoscopic variceal ligation or Nadolol/ Propranolol given to prevent deterioration of patient

PORTAL HYPERTENSION

00:25:54

DEFINITION

- Hepatic venous wedge pressure >5mm Hg
- **HVPG > 12 mmHg** → bleeding oesophageal varices
- IOC for Portal hypertension: **Doppler ultrasound**
- Earliest clinical finding of portal hypertension: Splenomegaly
- In liver cirrhosis → regression in size of liver due to fibrosis → resistance in vessels ↑ → pressure in portal vein ↑ → transmission to splenic vein → congestion in spleen → splenomegaly → transudation (**SAAG > 1.1g/dl**) of fluid in peritoneal cavity due to splanchnic blood vessels
- Development of Porto-systemic collaterals
- Dilated veins around the umbilicus (Caput medusae) → away from the umbilicus
- Portosystemic collaterals → pressure of portal veins transmitted to rectal veins → rectal varices
- Esophageal varices

PORTOSYSTEMIC COLLATERAL SITES

- Caput medusae
- Esophageal veins
- Rectal hemorrhoids
- Bare area of the liver
- Omental veins
- Retroperitoneal veins
- Lumbar veins

00:31:16



Caput medusae

CAUSES OF PORTAL HYPERTENSION

PRE HEPATIC	HEPATIC	POST HEPATIC
• Portal vein thrombosis • Splenic vein thrombosis • Banti syndrome (massive splenomegaly) ◦ Idiopathic ◦ Japan	• Pre- sinusoidal ◦ Schistosomiasis • Sinusoidal ◦ Cirrhosis due to any etiology • Post sinusoidal ◦ Veno-occlusive disease ◦ HSCT, Chemo, Radiation, Herbal tea ◦ Endemic ascites (Pyrrolizidine)	• Budd–Chiari Syndrome • IVC webs • Restrictive cardiomyopathy • Constrictive pericarditis (compliance will reduced)

Important Information

Bantu syndrome:

- Seen in Africa
- Consumption of locally brewed beer stored in iron utensils → iron toxicity

MANAGEMENT

00:37:10

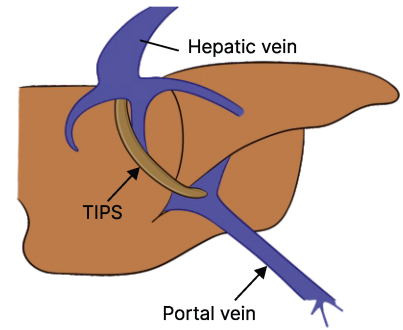
- TOC for non-bleeding esophageal varices: **Endoscopic variceal ligation**
- DOC for non-bleeding esophageal varices: **Nadolol/Propranolol**
- Bleeding esophageal varices with hemodynamic compromise:
 - 1st-line: Secure IV access (gray cannula) → Massive transfusion
→ 6 units PRBC : 6 units FFP : 1 unit SDP (1:1:0.25 ratio)
 - Target BP: **≥90/60 mmHg** (Permissive hypotension)
 - DOC for bleeding varices: **Octreotide infusion/Terlipressin**
 - TOC for bleeding varices with hemodynamic compromise: Endoscopic variceal ligation done after volume resuscitation



MANAGEMENT FOR RECURRENT VARICEAL BLEEDING PATIENTS

Report E/L: 1st

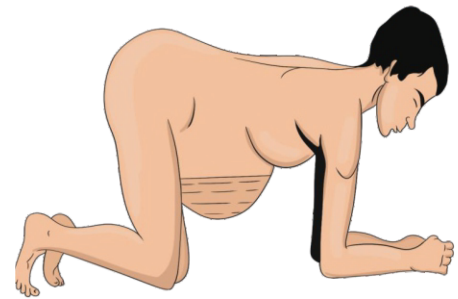
- TIPS (Transjugular intrahepatic portosystemic shunting):
 - Advantage: Portal blood enters systemic circulation directly, bypassing liver → hepatic encephalopathy & ammonia intoxication
 - Major source of ammonia in body → intestinal bacteria
 - Lactulose → remove faecal matter → ↓ ammonia production
- Shunt surgery (distal splenorenal shunt, Portocaval shunt)
- Orthotopic liver transplantation



ASCITIS

00:44:45

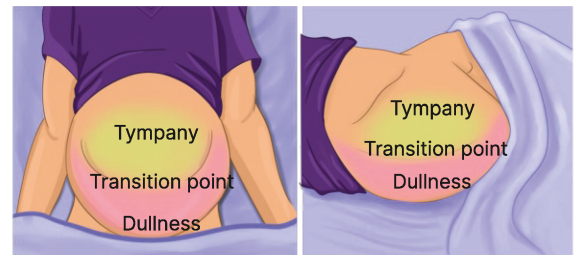
- Normal Peritoneal fluid: 0 mL
- Mid-cycle (ovulation in females): <10 mL
- IOC for ascites: USG abdomen
- Clinical detection requires >1500–2000 mL of fluid
- Puddle sign detects >120 mL fluid in peritoneal cavity
 - Patient positioned on all fours (elbows and knees)
 - Flick abdomen (percussion + auscultation)
 - Minimum fluid for detection: 1500–2000 mL



Puddle sign

CLINICAL SIGNS NOTED IN ASCITES

- Shifting dullness:
 - Fluid in peritoneal cavity shifts with position change → Gravity pulls fluid to dependent side → Dullness shifts accordingly
 - Fluid wave sign: False positive results in grand multipara
 - Prominent veins on anterior abdominal wall and shiny stretched abdominal wall
 - Everted umbilicus
 - Hepatic hydrothorax: Shortness of breath → due to
 - Limitation of movement of diaphragm
 - Transdiaphragmatic movement of fluid from peritoneal cavity to pleural cavity → hydrothorax → B/L pleural effusion
- Untreated ascites can progress
 - Massive ascites → tense ascites → abdominal compartment syndrome
 - Pressure from inferior mesenteric vein → compression the inferior mesenteric vein → gut gangrene → paracentesis needs to be done



Shifting Dullness

MANAGEMENT

00:51:07

- ↑ Aldosterone → ↑ Salt and water in body → Exaggerated urinary loss of potassium and hydrogen → metabolic alkalosis
 - Derangements in ascites: Metabolic alkalosis
- Salt-restricted diet
- Diuretics: Spironolactone + Furosemide
- Dose ↑: Spironolactone (400 mg) + Furosemide (160 mg/day)

- If gynaecomastia is troublesome, **Amiloride** can be used
- **Midodrine/Clonidine** (counteract splanchnic vasodilatation that is activating RAAS system)
 - Nitric oxide from liver → Splanchnic vasodilatation → ↓ Renal perfusion → RAAS activation → ↑ Aldosterone → Ascites in patient
 - Midodrine is also used in **hepatorenal syndrome**

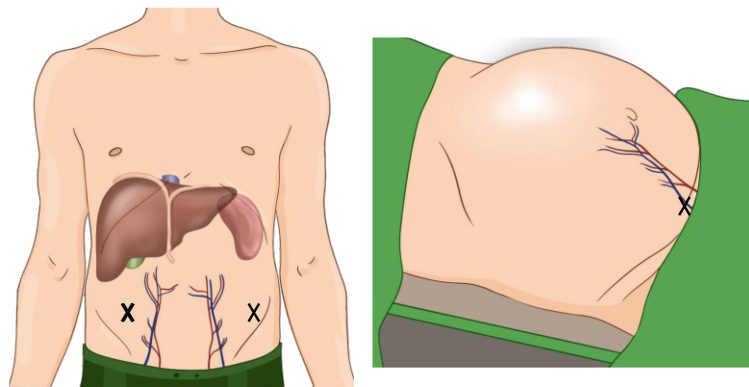
INDICATIONS FOR ASCITIC TAP

DIAGNOSTIC	THERAPEUTIC
• SBP: PMN > 250 cells/cumm • New onset ascites • Post admission in cirrhosis with ascites (done in <24 hours)	• Refractory ascites • Tense ascites causing SOB • Tense ascites causing compartment syndrome

PARACENTESIS

00:56:43

- Position: Head end → slightly elevated
- Site: **Left lower quadrant**
- Angle: needle introduced at **45°** in Z tracking mode



Site suggested for paracentesis

- Blood vessel which can be traumatised: Inferior epigastric artery → avoid putting needle inferiorly → **2cm** below the umbilicus → sub-umbilical site
- Flanks: **2-3 cm** superomedial to ASIS in the flanks
- ↑ Risk of caecal perforation on right side → Paracentesis recommended on left side

MANAGEMENT OF REFRACTORY ASCITES

- Large volume paracentesis/ low volume paracentesis with salt-free albumin
 - Objective: remove all fluid in a single sitting with albumin support
 - Salt-free albumin before the procedure → pulls fluid into intravascular compartment → ↓ risk of hypovolemic shock
 - Risk of recurrence remains
- TIPS → more effective at preventing fluid reaccumulation but ↑ risk of ammonia intoxication

SAC (SERUM ALBUMIN ASCITES GRADIENT)

TRANSUDATIVE		EXUDATIVE
Transudative etiology: >1.1 g/dl		Exudative etiology: <1.1 g/dl
Ascitic fluid protein <2.5 g/dL - Cirrhosis of any etiology - Late Budd-Chiari syndrome (obstruction + fibrosis)	Ascitic fluid protein >2.5 g/dL - CHF - Early Budd-Chiari syndrome - IVC obstruction - Veno-occlusive diseases	Causes: - Tuberculosis - Peritoneal carcinomatosis - Carcinoma stomach (adenocarcinoma) - Carcinoma ovary (serous cystadenocarcinoma) - Pancreatitis - Nephrotic syndrome - Biliary leak

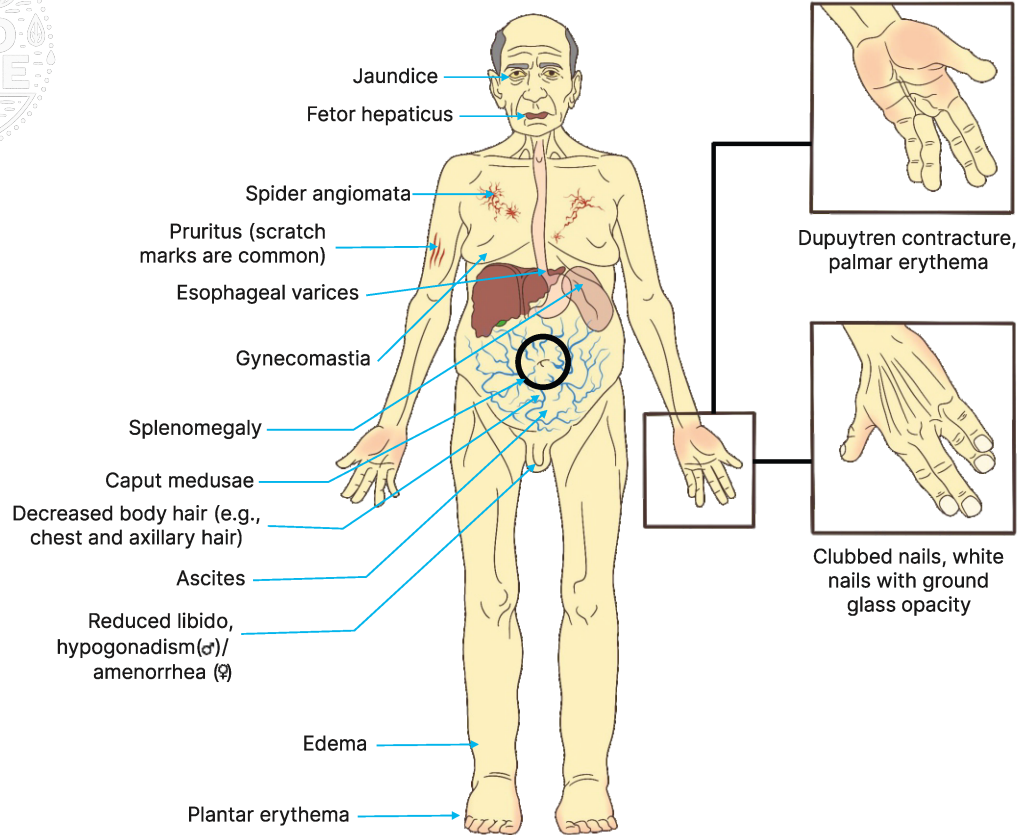
CASE SCENARIO

A 45-year-old alcoholic is admitted to the hospital for ascites with breathing trouble. 6 hours after admission, he develops fever with further increase in abdominal girth and worsening of dyspnoea. Pulse 105/min and BP 100/60 mmHg. Patient appears confused and disoriented, and diffuse abdominal tenderness is noted. Urgent paracentesis is done and shows 500 PMN cells/mm³. The fluid is sent for culture and Gram stain.

- Diagnosis: Spontaneous bacterial peritonitis (>250 PMN cells/mm³)
- Treatment: **Cefotaxime**
- Prevention: **Quinolones** (Norflox)
 - Cirrhosis patient with ascites → Perform paracentesis within 1st 24 hours
- Anticipated complication within 24 hours → **Hepatorenal syndrome**

SUMMARY OF FINDINGS SEEN IN DECOMPENSATED CIRRHOSIS

FINDINGS	NOTE
Bleeding esophageal varices	Octreotide and EVL
Refractory ascites	LVP with salt-free albumin and TIPS
SBP	Cefotaxime and norfloxacin for prophylaxis
Hepatorenal syndrome	Terlipressin/Octreotide and Albumin Midodrine to improve blood supply to kidney
Hepatopulmonary syndrome	OLT
Bleeding	FFP
Encephalopathy	Lactulose and Rifaximin (Bacteriostatic antibiotic-reduce the bacterial load)





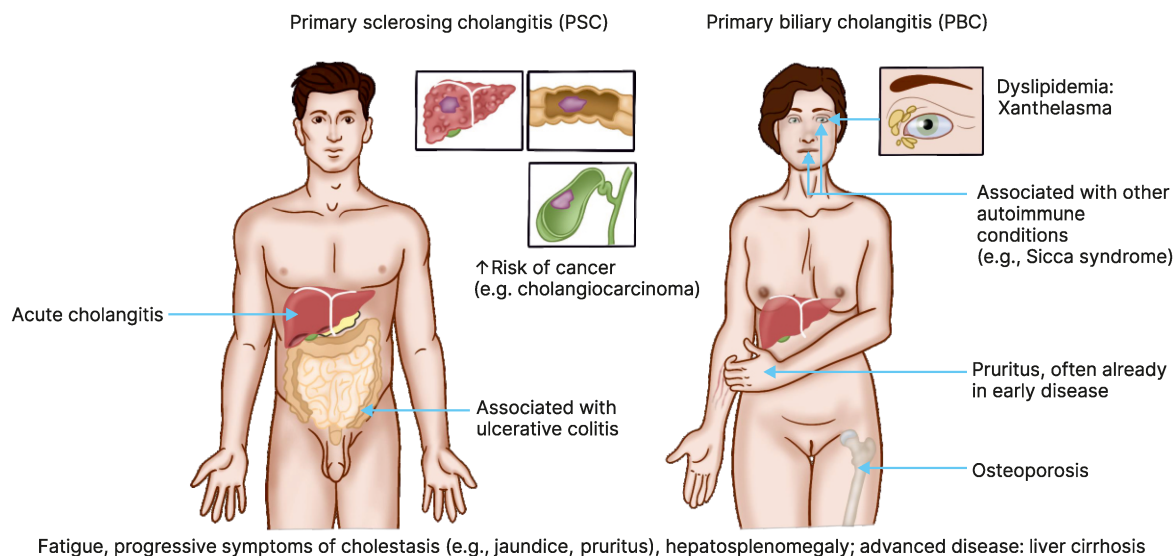
Q. A 35-year-old male patient was brought to the emergency department with RUQ pain and abdominal distention. He sometimes consumes beer. On examination, liver span is increased, and shifting dullness is noted. Diagnostic paracentesis shows SAAG >1.1 g/dl, ascitic fluid protein >2.5 gm/dl, S. bilirubin 4 mg/dl, SGOT and SGPT show moderate rise, Serum albumin 2.5 gm/dl, serum creatinine: 1mg/dl. USG shows enlarged caudate lobe of liver. Diagnosis?

- a. Spontaneous bacterial peritonitis
- b. Hepatorenal syndrome
- c. Budd-Chiari syndrome
- d. Alcoholic cirrhosis

Ans. (c)



IB, PBC AND PSC



PRIMARY BILIARY CHOLANGITIS (PRIMARY BILIARY CIRRHOSIS)

00:01:28

- Female patient
- Concomitant autoimmune disorder present:
 - Hashimoto's thyroiditis
 - Dry eyes and dry mouth : Sicca syndrome
 - Autoantibody : ANA (Anti-nuclear antibody)
 - Intrahepatic cholestasis

PATHOLOGY

- Caused by Anti-Mitochondrial antibody
- Intrahepatic lesion
- Non-supportive cholangiocyte damage

Important Information

- Anti-Microsomal Antibody - Hashimoto's thyroiditis
- Anti-Nuclear Antibody - SLE

FEATURES

- Cholestasis: c/o
 - Pruritus
 - Mustard yellow discoloration of urine
- On liver biopsy: Ductopenia → loss of intrahepatic interlobular bile ducts
 - Progressive ↑ in cholestasis
 - Granuloma formation
- Cirrhosis

CLINICAL FEATURES

- Main presentation of PBC : **Fatigue**
- Extensive pruritus followed by lichenification of skin
 - Due to ↑ activity of **μ-opioid receptors** and ↑ level of endogenous opioids like met-enkephalin due to impaired hepatic clearance
- Jaundice (conjugative) and hepatomegaly initially
- As liver begins to shrink-
 - Palmar erythema
 - Spider naevi
 - Temporal and proximal muscle wasting
- Portal hypertension- Ascites, caput medusae
- Impaired clearance of LDL cholesterol from blood due to damage in liver and cholangiocytes → LDL cholesterol deposition
 - Tendon- **Xanthoma**
 - Deposited in skin-**Xanthelasma**
- Steatorrhea with deficiency of Fat-soluble vitamins
- Osteopenia : Vitamin D3↓
- Hashimoto's thyroiditis
- Sicca syndrome : Dry eyes and dry mouth
- KF Ring

WORK-UP

00:07:30

- LFT:
 - Conjugated Bilirubin ↑ (fluctuating levels as it is an autoimmune disorder)
 - **Serum alkaline phosphatase** ↑ to 4 times the baseline value
- **Anti-Mitochondrial Antibody +**
 - Directed against multiple antigens : M₁-M₉
 - AMA profiling → Determine the antibody directed against individual mitochondrial antigens
 - ANA is present in 20-50% of the patients
- PT/INR↑
- Portal hypertension → spleen size ↑ → hypersplenism → **Platelet ↓**
- Deranged Lipid profile
- USG abdomen to rule out extrahepatic causes of obstructive jaundice
- Main IOC: **Liver biopsy** → shows granuloma around small interlobular bile ducts

MANAGEMENT

00:09:48

- UDCA (Ursodeoxycholic acid) : Promotes bile flow **Choleretic effects**
- Obeticholic acid
- Cholestyramine
- Antihistaminic
- Dronabinol/Naltrexone } For pruritus
- Plasmapheresis : Limited efficacy
- TOC : **Orthotopic liver transplantation**

PRIMARY SCLEROSING CHOLANGITIS

00:11:01

- Multi patient
- **Structure formation** in extrahepatic biliary pathway
 - Common bile duct, Common hepatic duct → cholestasis → features of **obstructive jaundice**
- Associated with **colitis** (ulcerative colitis)
 - Ulcerative colitis can have extraintestinal manifestations also
 - Hepatobiliary involvement of primary sclerosing cholangitis
 - Both of these disorders share common antibodies **p-ANCA**

CAUSE OF PRIMARY SCLEROSING CHOLANGITIS

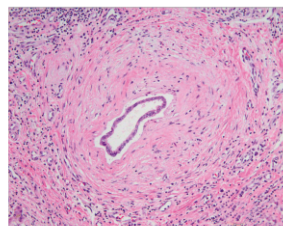
- Multifactorial
- Genetic predisposition
- Associated with **ulcerative colitis** (pre-malignant condition)
- ↑ risk of developing
 - Cholangiocarcinoma
 - Colon cancer

CLINICAL FEATURES

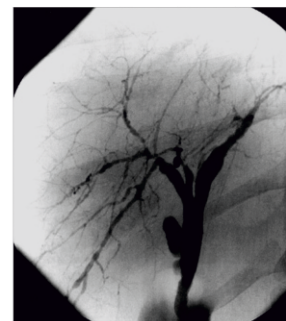
- Pruritus
- Icterus: **waxing and waning** course
- Steatorrhea
- 50% of cases are associated with Ulcerative Colitis

WORK UP

- LFT
 - Derangement of total serum Bilirubin
 - Conjugated bilirubin ↑
 - Serum Alkaline Phosphatase ↑ X 4 times of normal
- PT ↑
- Albumin ↓, as there is progressive cirrhosis
- **pANCA ⊕**: 65% of cases of PSC
- Imaging modality of choice and also IOC: **MRCP > ERCP**
- Liver biopsy: **onion skinning of bile ducts** (late manifestation)
- For initial demonstration of presence of multiple strictures: MRCP and ERCP



Concentric / "onion skin" fibrosis surrounding a bile duct in PSC



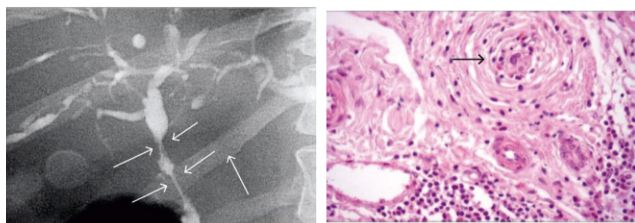
ERCP in PSC

MANAGEMENT

- UDCA
- ERCP with balloon dilatation of strictures
 - No role in intrahepatic involvement seen in Primary Biliary Cholangitis
- **Orthotopic liver transplant (OLT)**



Q. Which of the following is wrong about the condition shown below?



- a. Increased risk of ulcerative colitis
- b. Investigation of choice is ERCP
- c. Presents with cholestasis and pruritus
- d. High titres of anti-mitochondrial antibody

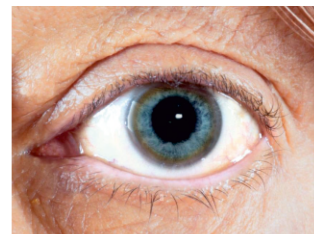
Ans. (d)

WILSON DISEASES AND HEMOCHROMATOSIS

KF RING (KAYSER-FLEISCHER RING)

00:00:10

- KF ring is also seen in: **Primary Biliary Cholangitis**
- Ceruloplasmin deficiency also seen in:
 - **Malnutrition**
 - **Aceruloplasminemia**
 - **Nephrotic syndrome**
- **Leipzig scoring system** used for diagnosis of Wilson diseases

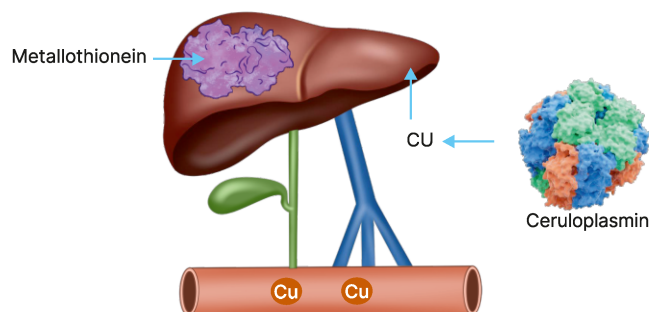


Kayser-Fleischer ring

WILSON DISEASES

00:01:59

- **Low ceruloplasmin** is the manifestation of Wilson diseases
- Total body copper : **50-100 mg**
- Consumption of copper: 2-5 mg/day
- Copper in food → Absorption from intestine → Normal in patients of Wilson diseases → Copper binds with **metallothionein** present in the liver → Excreted out of the liver
- **P-type ATPase** : Protein defective in Wilson diseases
 - Function- Hepatobiliary excretion of Copper
 - Copper in body is transported bound to ceruloplasmin but in case of defect of P-type ATPase → Copper comes into the liver but not excreted
- Wilson diseases: Defective hepatobiliary excretion of copper
- Defective gene: **ATP 7B**, chromosome 13, **Autosomal recessive** pattern of inheritance
- Similar Family history in siblings
- Basal ganglia involved in Wilson diseases: **Lenticular nucleus**
 - Hepatolenticular degeneration: Damage to basal ganglia → Neurological features



CLINICAL FEATURES

00:04:49

- Occurs in >5 years and <50 years of age

LIVER DAMAGE

- Usually in 1st decade)
- Damage to hepatocytes with extra copper
- Manifestations
 - Chronic hepatitis with viral markers
 - Decompensated cirrhosis with ascites and portal hypertension
- In Wilson diseases, serum copper ↓ and ↑ in tissues

Important Information

Genetic Diseases

• ATP7A gene

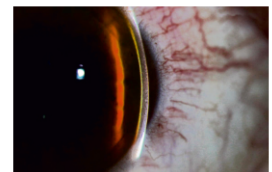
- X linked recessive
- Male: Mental retardation + kinky hair
- Copper deficiency

NEUROLOGICAL FEATURES

- Asymmetrical tremors in hands- **Wing beating tremors**
 - ↑ urinary copper
 - ↓ serum copper
 - Disease begins after 10 years of age
 - To elicit wing beating tremor: Pt is asked to bring both his hands at the level of nose → tremor noticed like beating of wings of bird
 - Low frequency high amplitude proximal tremor involving **Dentato-rubro- thalamic pathway**
- Dysarthria, drooling of saliva, dystonia
- Behavioral changes: Disinhibition, poor scholastic performance, impulsivity, emotional lability, migraine like headache (once the critical levels are attained, basal ganglia get involved)
- Hemolytic anemia
 - Concentration of copper inside RBC ↑↑ → RBC burst → Hemolytic anemia → Coomb's negative hemolytic anemia
- Joints: **Premature osteoarthritis**
- Classical manifestation: **Hepatolenticular** (first 10 years-hepato, after 10 years- lenticular nucleus), but not limited to lenticular nucleus → cerebellum can be involved- gait ataxia can be seen, balance issues

KF RING

- Appears by **10-12 years** of age but it may not be seen as initially the KF ring involves upper 2 mm of cornea and this part of cornea is not visible
- Subsequently may involve the lower part of cornea also → **12-15 years** of age → development of ring seen
- It can be:
 - Brownish green
 - Brownish golden
- ↑ circumferentially and involve whole of the cornea
- Slit lamp examination : **Descemet's membrane** involved
- Reversible with treatment: Chelator agent - **Trientine** → Reduces the level of copper in tissues → KF ring gradually resolve



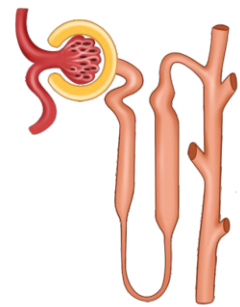
AZURE NAIL

- Starting part of the nail (lanula) → Bluish appearance (can also be seen in silver toxicity)





- Exposed with copper → RBC burst → Copper in urine via blood → Damage to glomerulus → Hematuria
- Copper also causes damage to proximal convoluted tubule → Fanconi syndrome
 - Aminoaciduria
 - Glucosuria
 - Phosphaturia
- Initially serum copper elevated → later serum copper ↓ → Deposited in tissues and lost in urine → no role of serum copper in diagnosis of condition



OTHER RARE MANIFESTATIONS

- Amenorrhea
- Cholelithiasis
- Nephrolithiasis

ORGANS INVOLVED IN WILSON DISEASES

LIVER	• Cirrhosis ◦ Initially compensated → Palmar erythema, spider angioma ◦ Subsequently decompensation occurs → Ascites and portal hypertension
RBC	• Coombs -ve hemolytic anemia
BASAL GANGLIA	Wing beating tremors
KIDNEY	Hematuria, Fanconi syndrome
NAILS	Azure nails
JOINTS	Osteoarthritis (knee, ankle joints)
EYE	KF ring
HEART	Conduction abnormalities

HEMACHROMATOSIS	WILSON DISEASES
• Heart involved: Restrictive cardiomyopathy • Type I diabetes as pancreas involved	• Does not cause restrictive cardiomyopathy • No involvement of pancreas

WORK UP

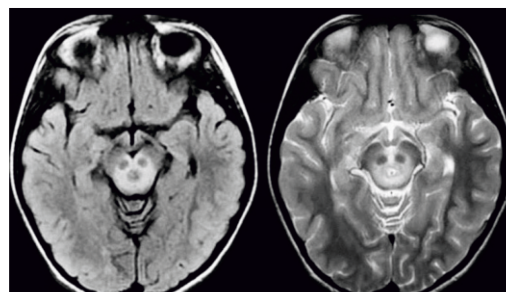
- LFT: elevated transaminases
- Serum ceruloplasmin <20mg/dl (screening) and 24-hour urinary copper value >100 mcg/dl as RBCs are destroyed
 - Combination of both of the above have diagnostic importance



Important information

Diagnosis of Hemochromatosis investigations done

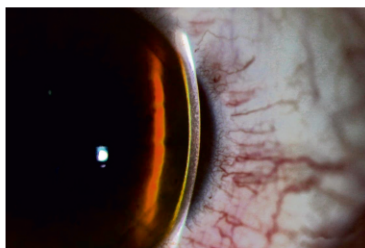
- Serum ferritin
- Transferrin saturation
- Serum copper
 - No diagnostic importance
 - Later part of the diseases copper ↓ but in initial part of diseases when hemolytic episodes occur, ↑ serum copper present
- IOC : **Liver biopsy** and estimation of hepatic copper content **>250 mcg/g** of dry weight of liver
- Stains used: Rhodanine, Rubeanic acid (unreliable)
 - Simply demonstrating presence of copper in hepatocytes (present in normal person also) is not sufficient for diagnosis
- Radiolabeled Cu-64: Fails to get incorporated into ceruloplasmin
- MRI head: **Giant face of panda appearance** (neurodegeneration at young age)
 - Tells about involvement of midbrain, basal ganglia, cerebellum of the patient can be involved



GIANT FACE OF PANDA APPEARANCE

Important Information

- **Panda sign** on gallium 67 scan : Sarcoidosis
- Panda sign due to trauma (**Raccoon eyes**) : Anterior cranial fossa
- Slit lamp biomicroscopic examination
- DNA testing for ATP7B gene/Haplotype testing in siblings
- **Nazer index**: To decide the treatment of the patient whether to go for medical treatment or orthoptic liver transplantation (mnemonic: **BIGT**)
 - Serum **bilirubin**
 - **INR**
 - **SGOT**



NAZER INDEX VALUE	
Medical treatment	<7
Orthoptic liver transplantation	>9



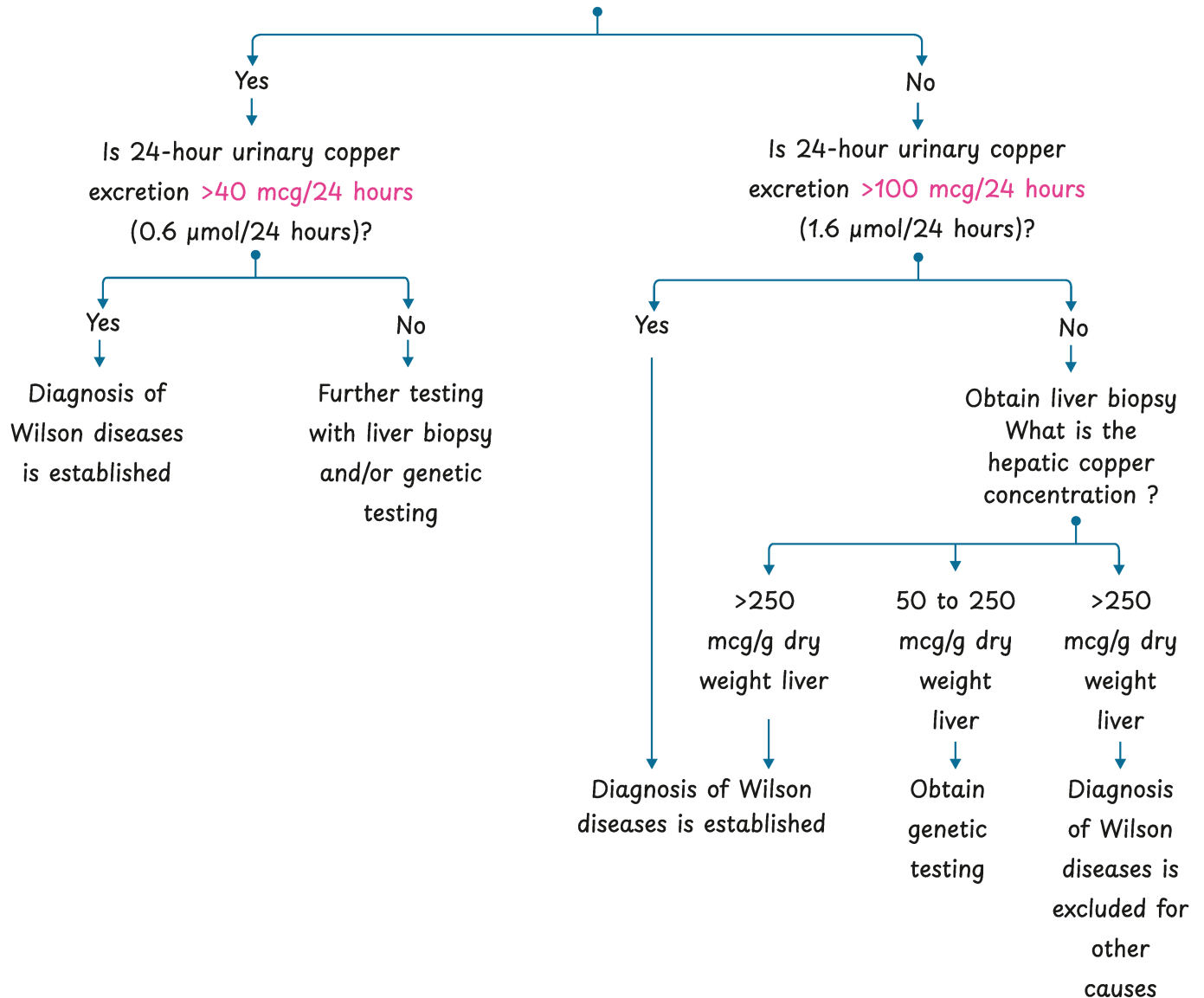
DIAGNOSTIC ALGORITHM

Symptomatic patients with KF rings

Review the following studies:

- Serum ceruloplasmin
- 24-hour urinary copper excretion

Is ceruloplasmin low (<20 mg/dl (200 mg/L) ?



LEIPZIG CRITERIA FOR DIAGNOSIS OF WILSON DISEASES

KF rings

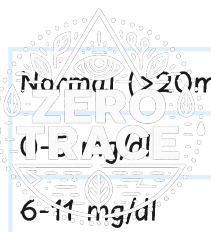
Present

2

Absent

0

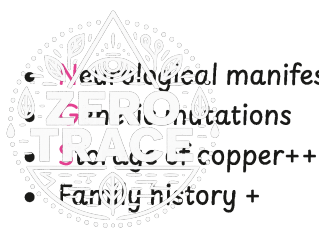
Serum ceruloplasmin



Normal ($>20\text{mg/dl}$)	0
(1-5 mg/dl)	3
6-11 mg/dl	2
11-20 mg/dl	1
24-h urinary copper (in the absence of acute hepatitis)	
$>100\text{ mcg}$	2
40-100	1
$<40\text{ mcg}$	0
Coomb's negative hemolytic anemia with liver disease	
Present	1
Absent	0
Mutational analysis	
On both chromosomes detected	4
On one chromosome detected	1
No mutation detected/test not done	0
Liver biopsy for histology S/O WD	
Orcein or rhodamine positive granules	1
Neurobehavioral symptoms	
Typical features on MRI brain	1
History of Wilson disease in a family member	
Sibling death from liver disease/neurological disease suggestive of WD	1
Total score	Evaluation
≥ 4 or more	Diagnosis established
3	Diagnosis possible, more tests needed

Mnemonic: **CU-RINGS**

- Ceruloplasmin ↓
- Urine copper ↑
- KF rings
- Hemolysis



- Neurological manifestations + MRI brain
- Genetic mutations
- Levels of copper++
- Family history +

MANAGEMENT

00:28:20

- 1st line drug/Chelator of choice: **Trientine** (D-penicillamine can cause damage to kidneys)
 - Trientine → Binds to copper and helps in excretion of copper via urine
- Maintenance: **Zinc acetate**
 - Competitive inhibitor of action of copper
 - MOA: Induces **metallothionine enterocytes** and copper is trapped in intestinal cells which are shed subsequently (GI shedding of copper rich enterocytes)
- CNS manifestations: **Tetrathiomolybdate**: Promotes urinary excretion of copper

HEMOCHROMATOSIS

00:31:14

- Total iron content in normal person : **3-4 gm**
- Total iron content in Hemochromatosis case : **20-25 gm**
- For treatment of hereditary hemochromatosis: Phlebotomy done
 - Blood from the body is removed
 - With 1 unit of blood: **250 gm of iron** removed → aggressive phlebotomy treatment needs to be done in patients of iron toxicity especially Hereditary hemochromatosis

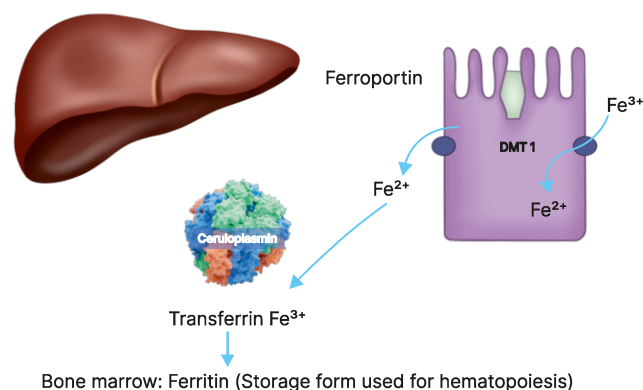
HEREDITARY HEMOCHROMATOSIS

- M/c symptoms in patients of Hemochromatosis: **Joint pain > skin pigmentation** (slate/grey color discoloration as local iron deposition causes excess melanin production)
- M/c organ involved in Hemochromatosis: **Liver**
- M/c cause of death in Hemochromatosis: Cardiac involvement → Restrictive cardiomyopathy → Features of **Congestive heart failure**
- Clinical features that persist in spite of treatment: Joint damage

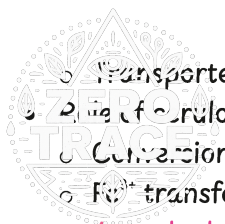
PHYSIOLOGY OF IRON ABSORPTION

00:34:01

- Luminal side of intestine:
 - Fe^{3+} and Fe^{2+} present
 - **DMT 1** (Divalent Metal Transporter 1): Responsible for assimilation of iron:
 - Iron moves inside the cell after being converted from Fe^{3+} to Fe^{2+} (Fe^{2+} in diet will be absorbed and Fe^{3+} in diet will be converted to Fe^{2+})
 - Fe^{2+} inside the enterocytes
- At the site of presence of blood vessels: Ferroportin
 - Fe^{2+} transported into the circulation
 - Fe^{2+} cannot bind to transferrin



Transport form of iron	Transferrin
Storage form of iron	Ferritin



- Transported in form of Fe^{3+}
- Role of Ceruloplasmin
- Conversion of Fe^{2+} to Fe^{3+} so that it can bind to transferrin
- Fe^{3+} transferrin → Bone marrow → Ferritin (storage form used for hematopoiesis)
- **Aceruloplasminemia** contributes to development of Hemochromatosis as Fe^{2+} is not converted to Fe^{3+} → Not able to bind to transferrin → Deposits randomly in tissues
- Regulator of entire cycle of iron assimilation : **Hepcidin**
 - In case of absence of Hepcidin → Unopposed working of cycle → Hereditary hemochromatosis

HEREDIATRY HEMOCHROMATOSIS

00:36:44

- Defect of **HFE gene**, C282Y homozygosity- Chromosome 6
- Hepcidin is absent causing unregulated iron absorption
- Aceruloplasminemia : Iron coming in cannot be assimilated into transferrin
 - Extra iron load in the body contribute to hemochromatosis

TYPES OF HEMOCHROMATOSIS

HEREDITARY	ACQUIRED/ SECONDARY
Type I • HFE gene 6p, AR, C282Y Homozygosity	Chronic liver diseases • Liver damage → Hepcidin ↓→ Unregulated process related damage • Causes of chronic liver diseases → Contribute to Hemochromatosis ○ Hepatitis C (M/c cause) ○ Hepatitis B ○ Alcohol induced damage Chronic haemolytic anaemia • Thalassemia → Recurrent blood transfusion → Liver damage in case iron chelator not given • Myelodysplastic syndrome ○ Bone marrow transplantation required → Pending bone marrow transplantation → Blood transfusion required • Aplastic anaemia → Pending bone marrow transplantation → Multiple blood transfusion • Any haematological conditions requiring recurrent transfusion for survival Sideroblastic anaemia • Due to Isoniazid → B6 responsive anaemia • Ringed sideroblast are iron rich RBC- Extra amount of iron contribute to hemochromatosis
Type II • Hepcidin/ Hemojuvelin mutation	
Type III • Transferrin receptor mutation	
Type IV • Ferroportin mutation	
Miscellaneous cause: Aceruloplasminemia	